Answer **all** the questions in the spaces provided

1 A ball falls vertically in air from rest. The variation with time t of distance d moved by the ball is shown in Fig. 1.1.

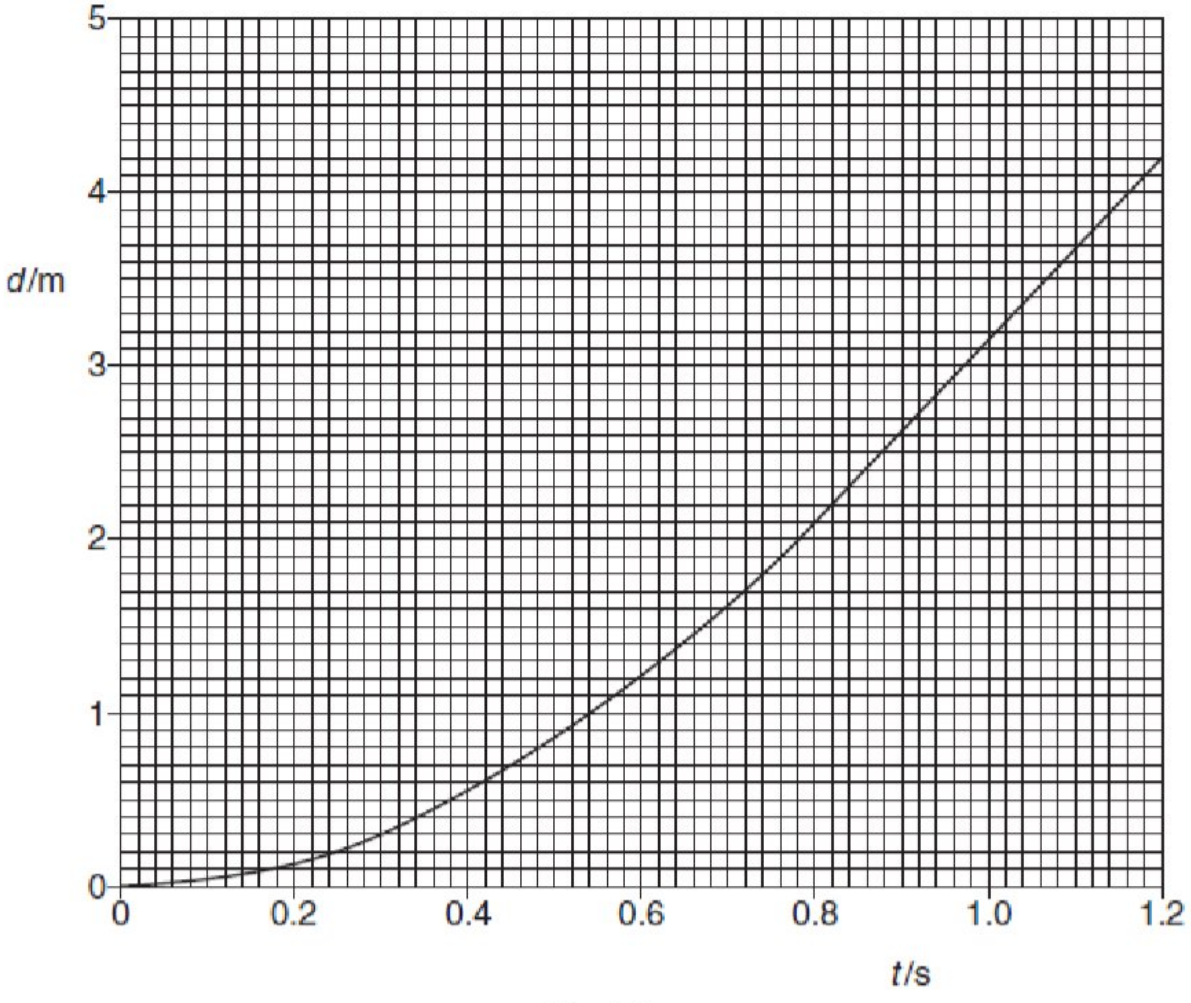


Fig. 1.1

(a) With reference to Fig. 1.1,

1. explain how it can be deduced that the ball is initially at rest,

.....

.....

..... [1]

2. explain, with reference to forces acting on the ball, why air resistance is not negligible.

.....

.....

..... [2]

(b) Use Fig. 1.1 to determine the speed of the ball at a time of 0.40 s after it has been released.

speed = ... m s^{-1} [2]

(c) The same ball is now tossed vertically into the air with velocity V to reach a maximum displacement of 2.0 m.

Sketch a graph on the axis below to show the variation with time t of the velocity v of the ball from when it is tossed to when it returns to its original position.

Mark the instant at which the magnitude of acceleration is g = 9.81 m s^{-2} as t_1 and the instant at which the ball returns to original position as t_2. No further calculations are required.

[3]



[Total: 8]